

Steady State Analysis of Nonlinear Dielectrics Filled Microwave Passive Devices

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Abstract—This letter presents the computation of high order harmonic components of the time-harmonic field exciting waveguide structures partially filled with nonlinear dielectrics. The approach presented employs the Harmonic Balance method to perform a multi-harmonic Finite Element analysis, and the solution of the nonlinear system is handled with a Picard iteration. Since the nonlinear loop is particularly time demanding, a domain decomposition method is here proposed to speed-up the system solution, restricting the iteration loop to small sub-domain.

Index Terms—harmonic balance, finite elements, domain decomposition, nonlinear dielectrics, waveguide filter

I. INTRODUCTION

TODAY's microwave and millimeter wave passive devices design require modeling of material nonlinear properties to accurately predict the electromagnetic behavior, especially when high power densities are involved and dielectric materials are used. A time-stepping approach to the solution of Maxwell's equations in such cases require considerable amount of simulation times. An alternative time-harmonic approach, based on the finite element (FE) solution of a partial differential equation (PDE) of time harmonic variable, provides the fields quantities at a given selected frequency. Unfortunately, a single frequency is not sufficient to capture all relevant phenomena, high order harmonics of the fields been generated from the interaction of the electromagnetic field with the nonlinear materials. Incorporating a larger set of harmonics in the FE simulation by using the Harmonic Balance Finite Element (HBFE) method [1], [2], also known as or Multi-Harmonic Finite Element method [3]–[6], provides a fast and accurate solution to the problem, as long as the relevant harmonics required to affordably represent the field is not excessive. The nonlinearity of the PDE is tackled by an iterative solution of the so-built large-scale coupled linear system, updating the system at each step with its new linearized approach, converging within a prescribed tolerance to the sought solution. Several techniques are reported in literature, such as the Newton-Raphson algorithm. To improve the computational timings, an algebraic domain decomposition technique is applied as proposed in [7], and significant improvements are achieved when the region occupied by nonlinear materials is rather small.

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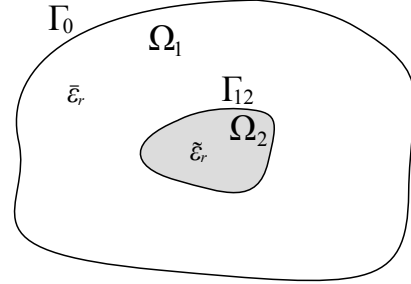


Fig. 1. Sketch of the domain $\Omega = \Omega_1 \cup \Omega_2$. increase font size

In this letter, we focus on the analysis of microwave devices filled with dielectrics presenting nonlinear scalar relative permittivity. In section II, we present the Harmonic Balance Finite Element method formulation for the solution of the vector Helmholtz equation for the electric field, within a domain comprising nonlinear dielectrics. In section III, simulation results of a broadband E-plane bandpass filter filled with Kerr-type material are shown, and performances of the algorithm are discussed.

II. FORMULATION

The formulation given in this section holds for the case of a generic electromagnetic problem defined on a closed domain as depicted in Fig. 2. The domain Ω , with boundary Γ_0 , can be either defined on $\mathbb{R}^2$ or $\mathbb{R}^3$. For the sake of simplicity, the examples presented in the next section are analyzed in $\mathbb{R}^2$, allowing for straightforward implementation. Ω is divided into a large part, Ω_1 , with linear scalar relative permittivity $\bar{\epsilon}_r(\mathbf{r})$ and a small part, Ω_2 , with a nonlinear relative permittivity $\tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}))$. The scalar permittivity in Ω can be expressed as

$$\epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \begin{cases} \bar{\epsilon}_r(\mathbf{r}), & \mathbf{r} \in \Omega_1 \\ \tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r} \in \Omega_2 \end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field pointed by $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator that describes the nonlinear behavior. The electric field satisfies, within Ω , the vector Helmholtz equation

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2 \epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \quad (2)$$

with suitable boundary conditions on Γ_0 . For the example shown in section III, Γ_0 is considered partly perfect electric conductor, partly a modal expansion of the field at the wave ports as developed in [8].

We proceed with the standard FE Galerkin framework: the domain Ω is discretized into a set of non overlapping finite elements and the electric field approximated as a linear combination of vector basis functions α_n

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^N E_n \alpha_n(\mathbf{r}) \quad (3)$$

Then by testing (2) over Ω with the weighting functions $\mathbf{W}_m$, chosen the same as the shape functions α_n , we obtain the following weak form:

$$\begin{aligned} \int_{\Omega} \nabla \times \mathbf{W}_m(\mathbf{r}) \cdot \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Omega \\ - k_0^2 \int_{\Omega} \mathbf{W}_m(\mathbf{r}) \cdot \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) d\Omega \\ + \oint_{\Gamma_0} \mathbf{W}_m(\mathbf{r}) \cdot \hat{n} \times \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Gamma = 0 \end{aligned} \quad (4)$$

We introduce the multi-harmonic dependence of the HBFE method, by approximating the electric field as a truncated Fourier series [4]

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{i=1}^M \mathbf{E}^{is}(\mathbf{r}) \sin(i\omega t) + \mathbf{E}^{ic}(\mathbf{r}) \cos(i\omega t) \quad (5)$$

where ω is the fundamental angular frequency. $\mathbf{E}_i^s$ and $\mathbf{E}_i^c$ are the harmonic complex amplitudes which expand the finite element coefficients E_n , which must individually satisfy (2). The scalar relative permittivity can also be approximated as a truncated Fourier series

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}, t) = \varepsilon_{r0} + \sum_{k=1}^K \varepsilon_r^{ks} \sin(k\omega t) + \varepsilon_r^{kc} \cos(k\omega t) \quad (6)$$

and the coefficients of the expansion are given by

$$\varepsilon_{r0} = \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) dt \quad (7)$$

$$\varepsilon_r^{ks} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \sin(k\omega t) dt \quad (8)$$

$$\varepsilon_r^{kc} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \cos(k\omega t) dt \quad (9)$$

with $k = 1 \dots K$. The orders M and K of approximation of the electric field and the permittivity, respectively, depend on the specific device analyzed and on the intensity of the incident field. These can be set by choosing an appropriate convergence criterion of the power associated to the harmonics of lower order as the number of multi-harmonic coefficients is increased.

We proceed substituting $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (4), then testing the equation with $\sin(i\omega t)$ and $\cos(i\omega t)$, $i = 1 \dots M$, and integrating over the time period of the fundamental $[0, \frac{2\pi}{\omega}]$ gives a large-scale linear system of the form:

$$\begin{bmatrix} a_{11}^{1s1s} & a_{11}^{1s1c} & \dots & a_{1N}^{1sMc} \\ a_{11}^{1c1s} & a_{11}^{1c1c} & \dots & a_{1N}^{1cMc} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1}^{Mc1s} & a_{N1}^{Mc1c} & \dots & a_{NN}^{McMc} \end{bmatrix} \begin{bmatrix} E_1^{1s} \\ E_1^{1c} \\ \vdots \\ E_N^{Mc} \end{bmatrix} = \begin{bmatrix} b_1^{1s} \\ b_1^{1c} \\ \vdots \\ b_N^{Mc} \end{bmatrix} \quad (10)$$

The product that occurs in the second integral of 4, between the approximated field and permittivity, induces coupling between the harmonic components. The order of the resulting HBFE system is of NM . In practical analyses, this number can become considerable, and as introduced previously, the solutions have to be computed iteratively, building at each step a new linearized system, until convergence is achieved. The chosen nonlinear solution is based on a Picard iteration: the permittivity is assumed initially to be the one in the absence of electric field:

$$[A_{([E]^0)}][E]^1 = [b_{([E]^0)}] \quad (11)$$

where $[E]^0$ is a null vector of unknowns. The solution of the first linear system of (12) provides expansion coefficients that will be used to compute a new value of the permittivity:

$$[A_{([E]^{q-1})}][E]^q = [b_{([E]^{q-1})}] \quad (12)$$

The process is repeated until the updated solution do not differ from the previous one, in the sense of the euclidean norm, more than a prescribed value.

The Picard iteration is not always stably convergent, especially for highly nonlinear materials and high impressed fields. Improvements can be achieved updating the system using fields solutions given by a proper linear combination of previous solutions [9].

To enhance the computational efficiency, a substructuring domain decomposition approach is used [7], [10]–[12]. The multiharmonic unknown vector $[E]$ defined for the HBFE method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2 , respectively. Unknowns belonging to the boundaries are placed in a third vector $[E^c]$. The HBFE system can be recast in the form:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M_{([E]^{q-1})}^{(2)}] & [P_{([E]^{q-1})}^{(2)}] \\ [Q^{(1)}] & [Q_{([E]^{q-1})}^{(2)}] & [C_{([E]^{q-1})}] \end{bmatrix} \begin{bmatrix} [E^{(1)}]^q \\ [E^{(2)}]^q \\ [E^c]^q \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b_{([E]^{q-1})}^{(2)}] \\ [b_{([E]^{q-1})}^{(c)}] \end{bmatrix} \quad (13)$$

where $[M^{(1)}]$ contains the FE coefficients of the linear system related to the unknowns $[E^{(1)}]$ fully internal for Ω_1 , thus linear, and $[M^{(2)}]$ contains coefficients for the nonlinear sub-domain Ω_2 . $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between the interior unknowns and the ones on the boundary Γ_{12} collected in $[E^c]$. Using a Gauss elimination-type strategy as the Schur complement $[S_{([E]^{q-1})}]$, the boundary unknowns $[E^c]^q$ of (13) are computed by solving the reduced system

$$[S_{([E]^{q-1})}][E^c]^q = [d]^{q-1} \quad (14)$$

where

$$\begin{aligned} [S_{([E]^{q-1})}] &= [C_{([E]^{q-1})}] - [Q^{(1)}][M^{(1)}]^{-1}[P^{(1)}] \\ &\quad - [Q_{([E]^{q-1})}^{(2)}][M_{([E]^{q-1})}^{(1)}]^{-1}[P_{([E]^{q-1})}^{(1)}] \end{aligned} \quad (15)$$

and

$$\begin{aligned} [d]^k &= [b_{([E]^{q-1})}^{(c)}] - [Q^{(1)}][M^{(1)}]^{-1}[P^{(1)}] \\ &\quad - [Q_{([E]^{q-1})}^{(2)}][M_{([E]^{q-1})}^{(1)}]^{-1}[P_{([E]^{q-1})}^{(1)}] \end{aligned} \quad (16)$$

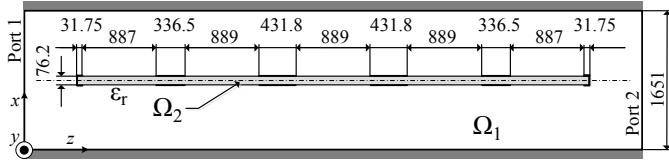


Fig. 2. Millimeter wave filter. increase font size

The difference operators in (15) and (16) are to be intended as the assembling of local matrices in a global one as performed by global FE matrix assembly [11]. Once $[E^{(c)}]^q$ is known, local solutions are given by

$$[E^{(1)}]^{q-1} = [M^{(1)}]^{-1} \left([b^{(1)}] - [P^{(1)}][E^{(c)}]^q \right) \quad (17)$$

$$[E^{(2)}]^{q-1} = [M_{([E]^{q-1})}^{(2)}]^{-1} \left([b_{([E]^{q-1})}^{(2)}] - [P_{([E]^{q-1})}^{(2)}][E^{(c)}]^q \right) \quad (18)$$

In order to solve the HBFE system and compute the steady state field, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 are to be updated. The efficiency of this DD technique relies on the fragmentation of the matrices to invert. When the number of coefficients related to nonlinear media is small, as usually happens for microwave devices where the domain of nonlinear media has reduced extension, noticeable improvements are achieved.

III. NUMERICAL RESULTS

In this section, we analyze the microwave filter sketched in Fig. 2. The dielectric material [13] (Yatsyk2006)

IV. CONCLUSION

The conclusion goes here.

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